

Phase 2 of Modern Control Project Control Design, Observer, and Robustness Analysis for the CSTR Reactor

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Abstract

In Phase 1, the nonlinear model of a continuous stirred-tank reactor (CSTR) was derived, linearized, and structurally analyzed. Building upon that linear model, this phase addresses advanced control system design. First, a statefeedback controller is designed and its domain of attraction on the nonlinear plant is mapped. The invariance of the transfer function under different realizations is demonstrated. A servo controller with integral action is then implemented to handle reference tracking and disturbance rejection, revealing the integrator windup phenomenon. To estimate unmeasured states, Luenberger and PI observers are developed, and their speedvs.noise tradeoff is studied. The combination of observer and controller is validated. Actuator saturation is investigated, and an antiwindup mechanism is employed. Finally, robustness against a 30 % increase in the inlet flow rate is confirmed. All simulations are carried out in MATLAB (total runtime ≈ 70 s).

1 Introduction and Review of Phase 1

The CSTR with an exothermic reaction and jacket cooling is modeled by the dimensionless statespace equations:

$$\frac{dx_2}{d\tau} = \frac{x_{60}}{x_1}(x_{20} - x_2) - c_0 x_2 e^{-1/x_3}, \quad (1)$$

$$\frac{dx_3}{d\tau} = \frac{x_{60}}{x_1}(x_{30} - x_3) + c_1 x_2 e^{-1/x_3} - \frac{c_2 c_3 x_5 (x_3 - x_{40})}{x_1 (c_3 x_5 + c_4)}, \quad (2)$$

where $x = [x_2 \ x_3]^T$ (dimensionless concentration and temperature) and $u = x_5$ (dimensionless cooling water flow rate). The equilibrium point at $u_{\text{nom}} = 1$ was found to be

$$x_{ss} = \begin{bmatrix} 1.941355 \\ 0.035537 \end{bmatrix}$$

with the corresponding linearized matrices

$$A = \begin{bmatrix} -1.0512 & -78.7580 \\ 0.0007 & -1.9024 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -7.87 \times 10^{-4} \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

The dimensionless parameters computed in Phase 1 are:

$$c_0 = 8.52106 \times 10^{10}, \quad c_1 = 1.10906 \times 10^9, \quad c_2 = 25.1524, \quad c_3 = 20.0071, \quad c_4 = 241.0782, \\ x_{20} = 2.040816, \quad x_{30} = 0.035095, \quad x_{40} = 0.035095, \quad x_{60} = 1, \quad x_1 = 1.$$

2 State Feedback Control and Domain of Attraction

The desired closedloop poles are chosen as $\{-3.0, -4.5\}$ to obtain a fast, welldamped response. Using MATLABs `place` command, the statefeedback gain is

$$K = \begin{bmatrix} 107.5446 & -5776.8742 \end{bmatrix}.$$

The control law $u = u_{\text{nom}} - K(x - x_{ss})$ is applied to the original nonlinear model (1)–(2). Simulations with five initial conditions progressively farther from the equilibrium (Fig. 1) show that the linear controller can stabilize the plant only within a limited region. A grid search over $x_2 \in [0.6, 3.2]$ and $x_3 \in [0.015, 0.10]$ identifies the approximate domain of attraction (Fig. 2): the green region contains initial states that converge to x_{ss} , while the red region leads to instability.

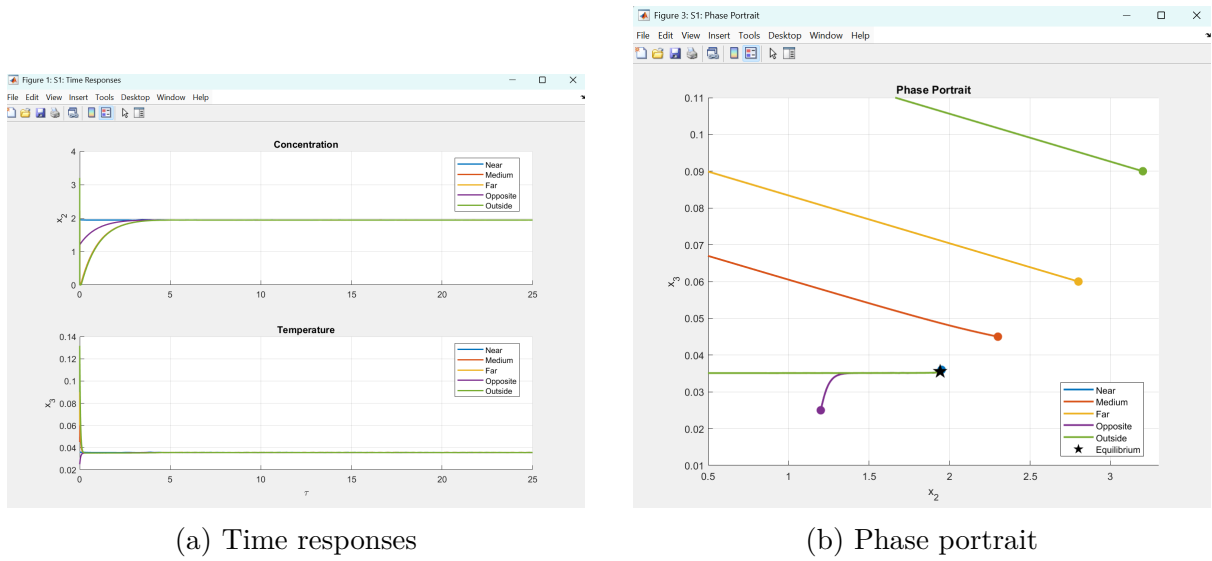


Figure 1: Response of the nonlinear system with statefeedback controller for different initial conditions.

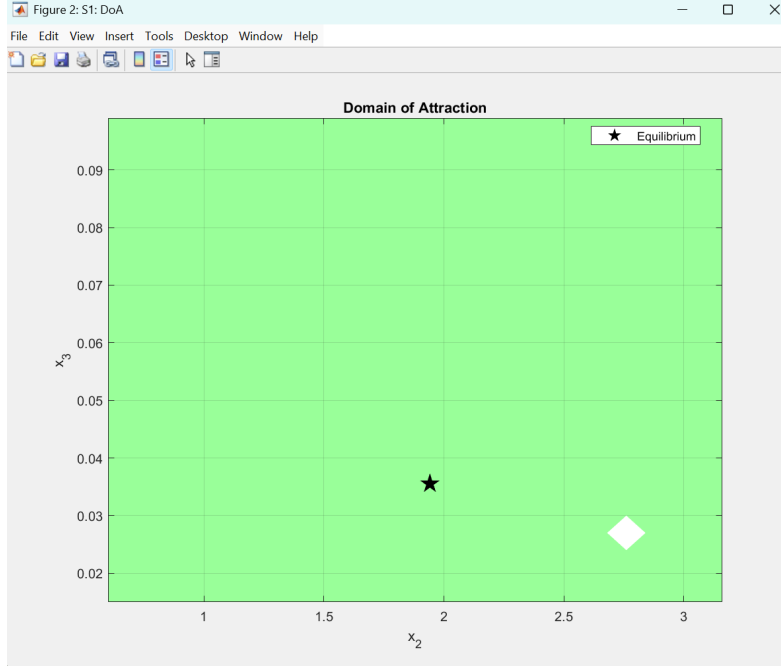


Figure 2: Estimated domain of attraction: green (stable), red (unstable).

3 Comparison of Different Realizations

The transfer function of the linearized plant is

$$G(s) = \frac{-7.870 \times 10^{-4} s - 8.273 \times 10^{-4}}{s^2 + 2.9536 s + 2.0549}.$$

The following realizations are constructed (numerical values rounded to four decimals):

- **Original:** (A, B, C) as above.
- **Controller canonical form:**

$$A_{ccf} = \begin{bmatrix} 0 & 1 \\ -2.0549 & -2.9536 \end{bmatrix}, B_{ccf} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C_{ccf} = \begin{bmatrix} -8.273 \times 10^{-4} & -7.870 \times 10^{-4} \end{bmatrix}.$$

- **Observer canonical form (dual):**

$$A_{ocf} = \begin{bmatrix} 0 & -2.0549 \\ 1 & -2.9536 \end{bmatrix}, B_{ocf} = \begin{bmatrix} -8.273 \times 10^{-4} \\ -7.870 \times 10^{-4} \end{bmatrix}, C_{ocf} = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

- **Alternative controller form (topcompanion):**

$$A_{ccf2} = \begin{bmatrix} -2.9536 & -2.0549 \\ 1 & 0 \end{bmatrix}, B_{ccf2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, C_{ccf2} = \begin{bmatrix} -7.870 \times 10^{-4} & -8.273 \times 10^{-4} \end{bmatrix}.$$

- **Alternative observer form:**

$$A_{ocf2} = \begin{bmatrix} -2.9536 & 1 \\ -2.0549 & 0 \end{bmatrix}, B_{ocf2} = \begin{bmatrix} -7.870 \times 10^{-4} \\ -8.273 \times 10^{-4} \end{bmatrix}, C_{ocf2} = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

- **Jordan form** (eigenvalues $-1.1218, -1.8318$):

$$J = \begin{bmatrix} -1.1218 & 0 \\ 0 & -1.8318 \end{bmatrix}, B_J = \begin{bmatrix} 0.0873 \\ -0.0873 \end{bmatrix}, C_J = \begin{bmatrix} 0.0009 & 0.0099 \end{bmatrix}.$$

For every realization, `ss2tf` returns exactly the same numerator and denominator, confirming that the inputoutput map is invariant under similarity transformation (Table 1). The step responses in Fig. 3 illustrate that, while the output y coincides for all forms, the state trajectories differ markedly.

Table 1: Transfer function coefficients for all realizations.

Realization	Numerator	Denominator
Original	$0s^2 - 7.870 \times 10^{-4}s - 8.273 \times 10^{-4}$	$s^2 + 2.9536s + 2.0549$
Controller CF	identical	
Observer CF	identical	
Controller CF (alt)	identical	
Observer CF (alt)	identical	
Jordan	identical	

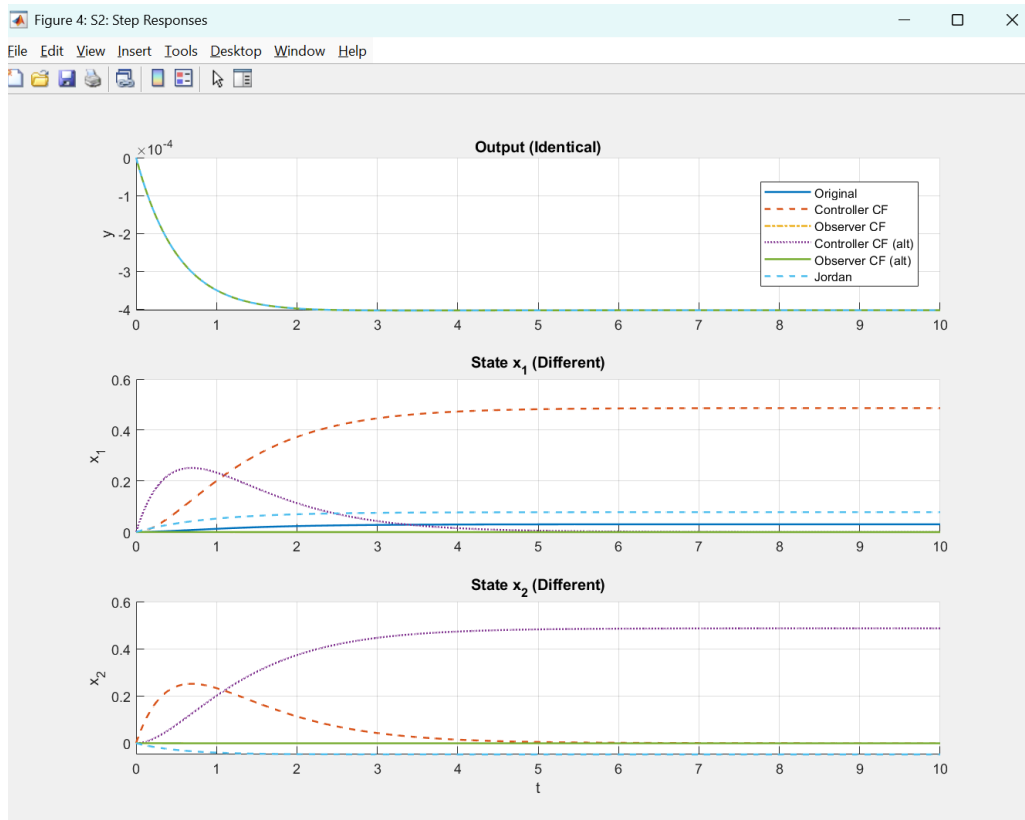


Figure 3: Step responses of different realizations: output y identical, states x_1, x_2 differ.

4 Servo Control with Integral Action

To achieve zero steadystate error under step references and disturbances, an integrator state is appended:

$$\dot{\xi} = r - y = r - Cx.$$

The augmented system becomes

$$\dot{x}_a = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix} x_a + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r, \quad x_a = \begin{bmatrix} x \\ \xi \end{bmatrix}.$$

Placing the closedloop poles at $\{-4, -5, -1.2\}$ yields the augmented gain

$$K_a = [K_s \ K_i] = \begin{bmatrix} -27.4819 & -9207.6239 & 29010.2290 \end{bmatrix},$$

so that $u = u_{\text{nom}} - K_s x - K_i \xi$. Figure 4 compares the classical state feedback (K) and the servo controller when a reference step $\delta r = 5 \times 10^{-4}$ occurs at $\tau = 2$ and a disturbance $d = 0.10$ enters at $\tau = 15$. The classical controller leaves a steadystate error ($\approx 7.14 \times 10^{-6}$), while the servo controller drives the tracking error to zero in theory. In practice, however, its mean absolute error after 35 s is larger (4.98×10^{-4}) due to integrator windup (see Fig. 5), a problem addressed in Section 7.

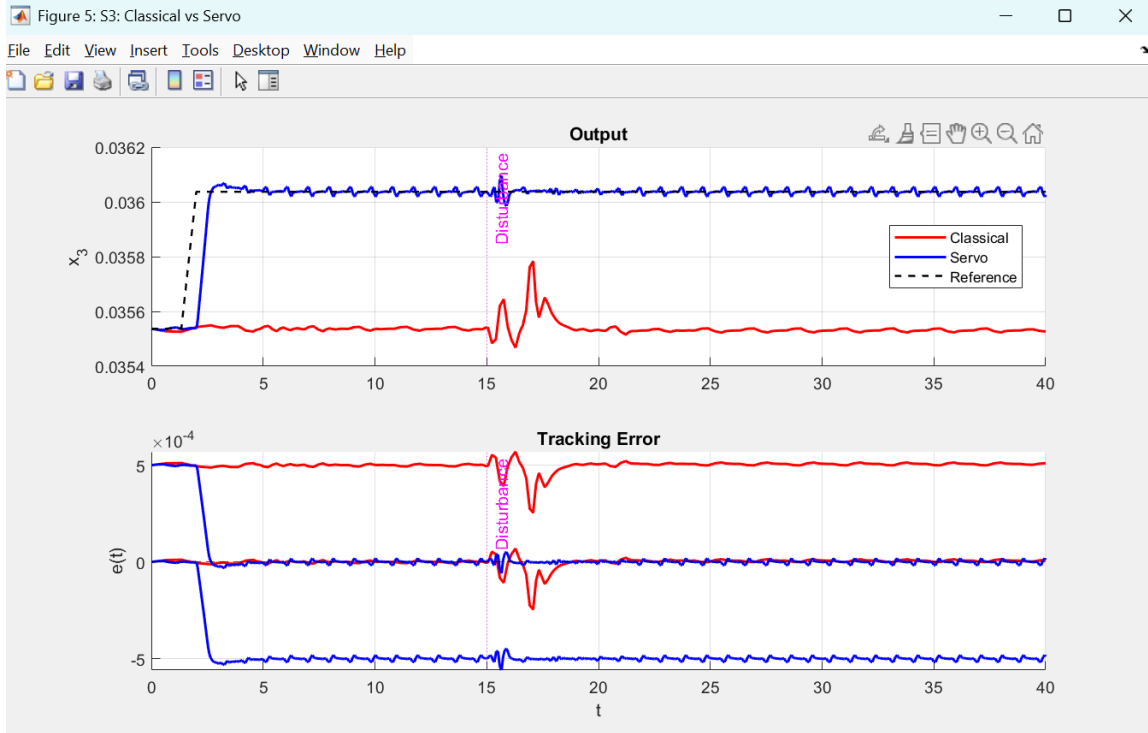


Figure 4: Classical vs. servo control under reference change and disturbance.

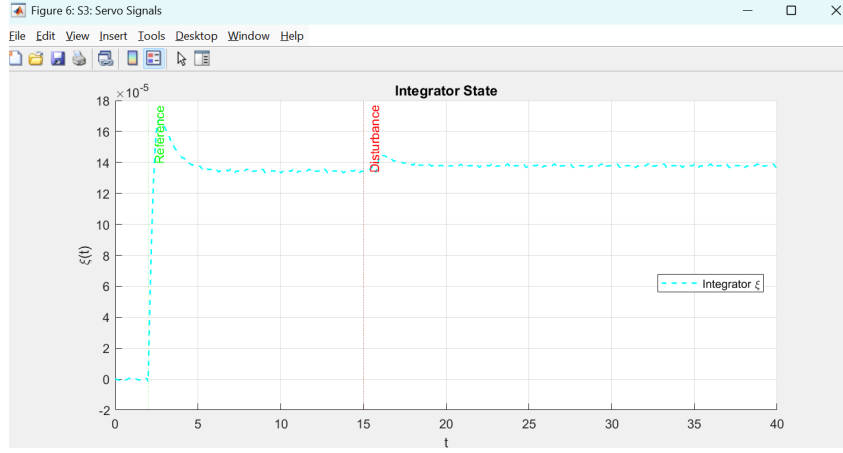


Figure 5: Integrator state ξ growing unbounded (windup).

5 Observer Design

5.1 Luenberger Observer

The Luenberger observer is

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}), \quad L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}.$$

Choosing observer poles faster than the controller (-9 and -12) gives the desired characteristic polynomial

$$\Delta_o(s) = (s + 9)(s + 12) = s^2 + 21s + 108.$$

The observers characteristic equation is

$$\begin{aligned} |sI - (A - LC)| &= \begin{vmatrix} s + 1.0512 & 78.758 + l_1 \\ -0.0007 & s + 1.9024 + l_2 \end{vmatrix} \\ &= s^2 + (2.9536 + l_2)s + 2.0549 + 1.0512l_2 + 0.0007l_1. \end{aligned}$$

Equating coefficients yields

$$l_2 = 18.0464, \quad l_1 = 1.2425 \times 10^5.$$

The very large l_1 arises because the (1,2) entry of A (-78.758) is much larger than the (2,1) entry (0.0007), making x_1 weakly observable. Despite this, the Luenberger observer works; the final estimation error after a disturbance at $\tau = 5$ is

$$x_1 \text{ error} = 2.82 \times 10^{-2}, \quad x_2 \text{ error} = -2.37 \times 10^{-7}.$$

5.2 PI Observer (Integral Observer)

To eliminate the steadystate estimation error, an integrator is added:

$$\dot{z} = y - C\hat{x}, \quad \dot{\hat{x}} = A\hat{x} + Bu + L_p(y - C\hat{x}) + L_i z.$$

The augmented observer dynamics are

$$\begin{bmatrix} \dot{\hat{x}} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A - L_p C & L_i \\ -C & 0 \end{bmatrix} \begin{bmatrix} \hat{x} \\ z \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} L_p \\ 1 \end{bmatrix} y.$$

Placing the augmented poles at $\{-1, -9, -12\}$ yields the target polynomial

$$(s + 1)(s + 9)(s + 12) = s^3 + 22s^2 + 129s + 108.$$

The determinant of the augmented matrix is

$$\begin{aligned} \begin{vmatrix} s + 1.0512 & 78.758 + l_1 & -l_3 \\ -0.0007 & s + 1.9024 + l_2 & -l_4 \\ 0 & 1 & s \end{vmatrix} &= s(s^2 + (2.9536 + l_2)s + 2.0549 + 1.0512l_2 + 0.0007l_1) \\ &\quad + (l_4s + 1.0512l_4 + 0.0007l_3) \\ &= s^3 + (2.9536 + l_2)s^2 + (2.0549 + 1.0512l_2 + 0.0007l_1 + l_4)s \\ &\quad + 1.0512l_4 + 0.0007l_3. \end{aligned}$$

Because of the degree of freedom, we set $l_1 = 0$ to avoid an excessively large gain. Equating coefficients:

$$\begin{aligned} 2.9536 + l_2 &= 22 &\Rightarrow l_2 &= 19.0464, \\ 2.0549 + 1.0512l_2 + l_4 &= 129 &\Rightarrow l_4 &= 129 - 2.0549 - 1.0512 \times 19.0464 = 106.923, \\ 1.0512l_4 + 0.0007l_3 &= 108 &\Rightarrow l_3 &= \frac{108 - 1.0512 \times 106.923}{0.0007} = -6282.869. \end{aligned}$$

Thus the PI observer gains are

$$L_p = \begin{bmatrix} 0 \\ 19.0464 \end{bmatrix}, \quad L_i = \begin{bmatrix} -6282.869 \\ 106.923 \end{bmatrix}.$$

The eigenvalues of the augmented matrix are exactly $\{-1, -12.0002, -8.9998\}$. After applying the PI observer, the final estimation error becomes practically zero (to within numerical precision).

5.3 Observer Speed Comparison

To study the tradeoff between speed and noise sensitivity, the PI observer is redesigned with three different pole locations, and measurement noise of power 10^{-12} is added. The following sets of gains are used:

- **Slow** (poles $\{-0.1, -0.9, -1.2\}$):

$$L_p = \begin{bmatrix} 0 \\ -0.7336 \end{bmatrix}, \quad L_i = \begin{bmatrix} 113.31 \\ 0.02728 \end{bmatrix}.$$

- **Mid** (poles $\{-1, -9, -12\}$): as derived above.
- **Fast** (poles $\{-10, -90, -120\}$):

$$L_p = \begin{bmatrix} 0 \\ 217.0464 \end{bmatrix}, \quad L_i = \begin{bmatrix} -3.576 \times 10^6 \\ 12669.785 \end{bmatrix}.$$

- **Very slow** (poles $\{-0.01, -0.09, -0.12\}$):

$$L_p = \begin{bmatrix} 0 \\ -2.7336 \end{bmatrix}, \quad L_i = \begin{bmatrix} -1248.61 \\ 0.8356 \end{bmatrix}.$$

The simulations (Fig. 6) reveal that slow observers filter noise well but cannot track the first state quickly; fast observers inject large noise into the control signal and cause oscillations. The middle setting $\{0.1, 0.9, 1.2\}$ offers a good compromise for this application.

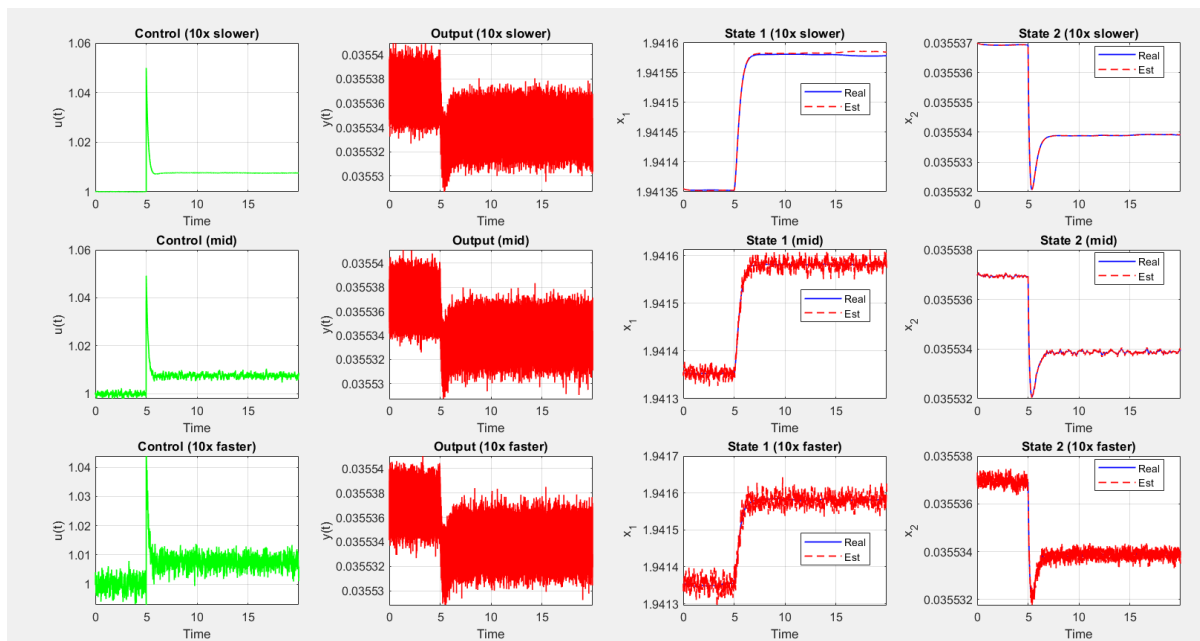


Figure 6: Observer speed comparison: control signal, output, and state estimates.

6 ObserverBased Control vs. Ideal State Feedback

When the estimated states from the slow PI observer are used for feedback, the controller must compensate for an initial estimation error of $[0.001, 0.0004]^T$. The observerbased system eventually converges to the same steady state as the ideal case, but with a transient overshoot and undershoot in the output (Fig. 7). Since the error is in the milliKelvin range, the transient is acceptable.

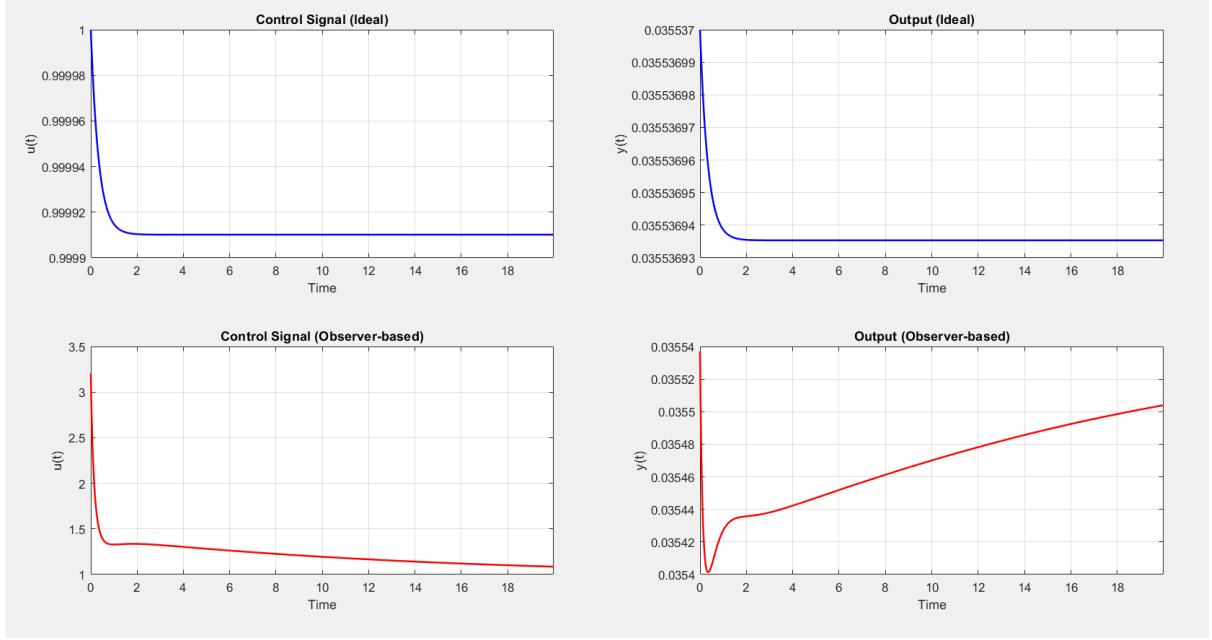


Figure 7: Ideal vs. observerbased control: control signal and output.

7 Input Saturation and AntiWindup

The control signal u (coolant flow rate) is physically bounded between 0 and 2 (twice the nominal value). To investigate the effect of saturation, the controller and observer gains are scaled by factors $\{1, 1.5, \dots, 5\}$, and the system is simulated with a disturbance at $\tau = 5$. Without antiwindup (Fig. 8), for scale factors ≥ 2 , windup becomes evident; at factor 5, the control signal oscillates between 0 and 2 (bangbang), and the output diverges.

A simple antiwindup strategy is implemented: the integrator state z is frozen ($\dot{z} = 0$) whenever the unsaturated control leaves the admissible interval. With this measure (Fig. 9), the system remains stable up to scale factor 4; at factor 5, windup is only partially mitigated, indicating the limits of actuator authority.

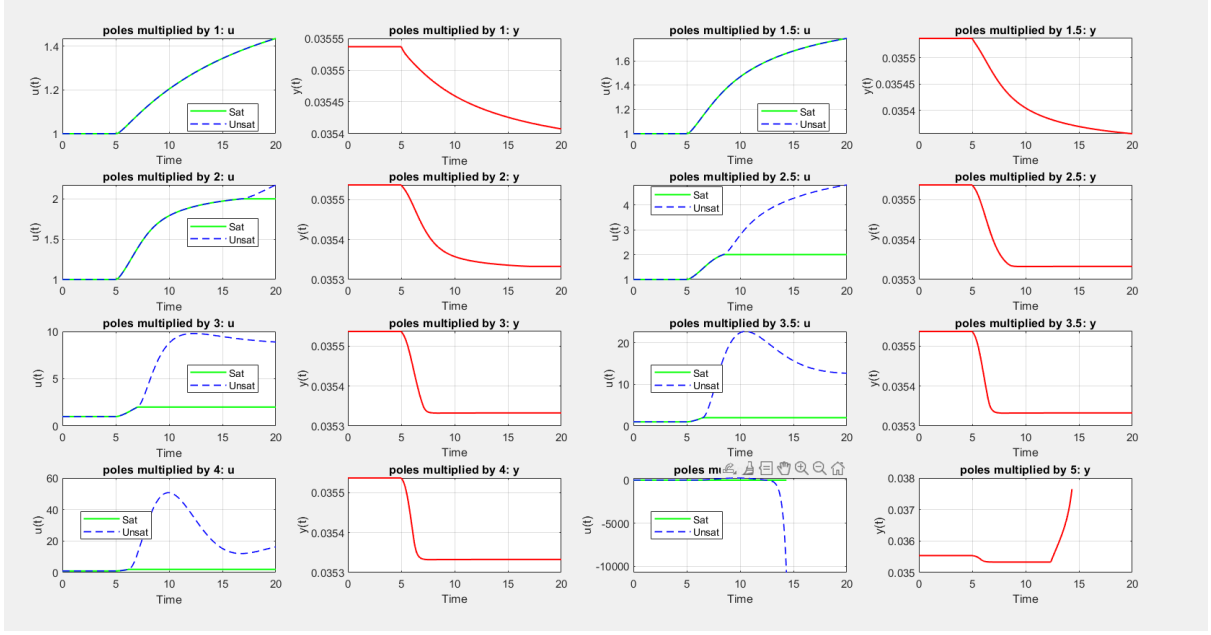


Figure 8: Saturation effect without antiwindup.

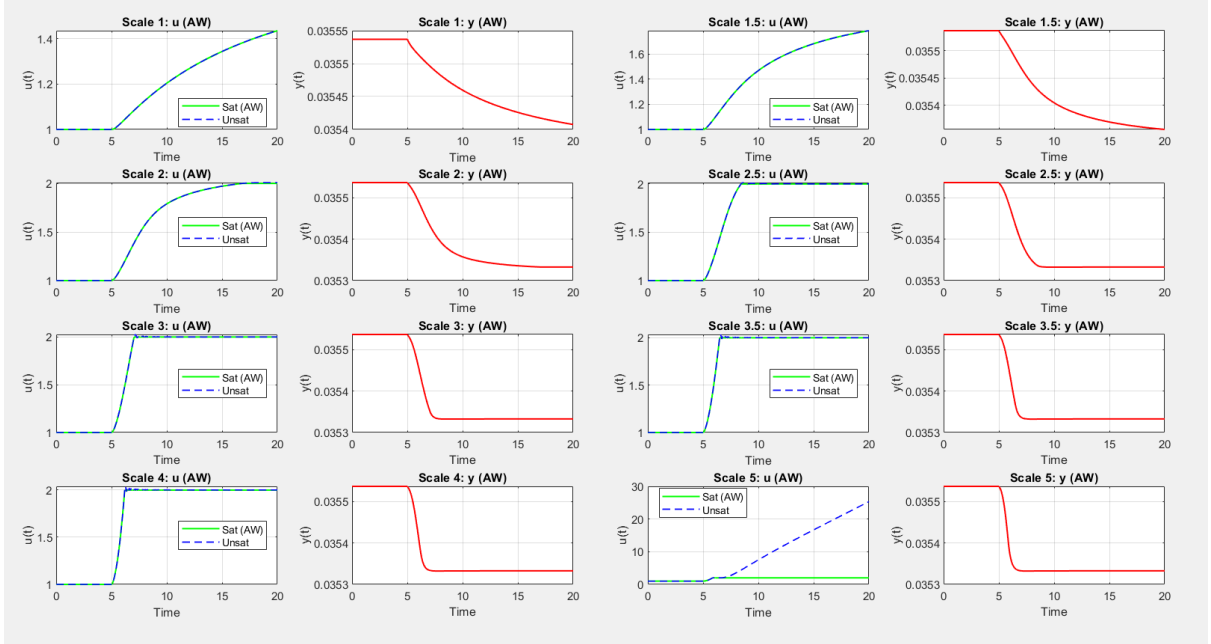


Figure 9: Performance with antiwindup.

8 Robustness to Parameter Uncertainty

A critical source of uncertainty is the inlet flow rate F_{os} . Increasing it by 30% changes all dimensionless parameters:

$$c_0 = \frac{V_s \alpha}{F_{os}}, \quad c_1 = \frac{V_s \alpha R C_{aor}}{F_{os} \rho C_p}, \quad c_2 = \frac{U A_{ht}}{\rho C_p F_{os}}, \quad c_3 = \frac{V_s F_{js}}{F_{os} V_j}, \quad c_4 = \frac{\rho C_p V_s}{\rho_j C_{pj} V_j} c_2.$$

The nominal controller/observer (scale factor 2) is applied to the plant with these perturbed parameters. Even without saturation, the output estimation error remains small

(-7.43×10^{-7}) . With saturation, the error is slightly larger (-6.65×10^{-6}) , but the system remains stable and tracks the desired operating point (Fig. 10). This confirms the robustness of the designed feedback/observer structure.

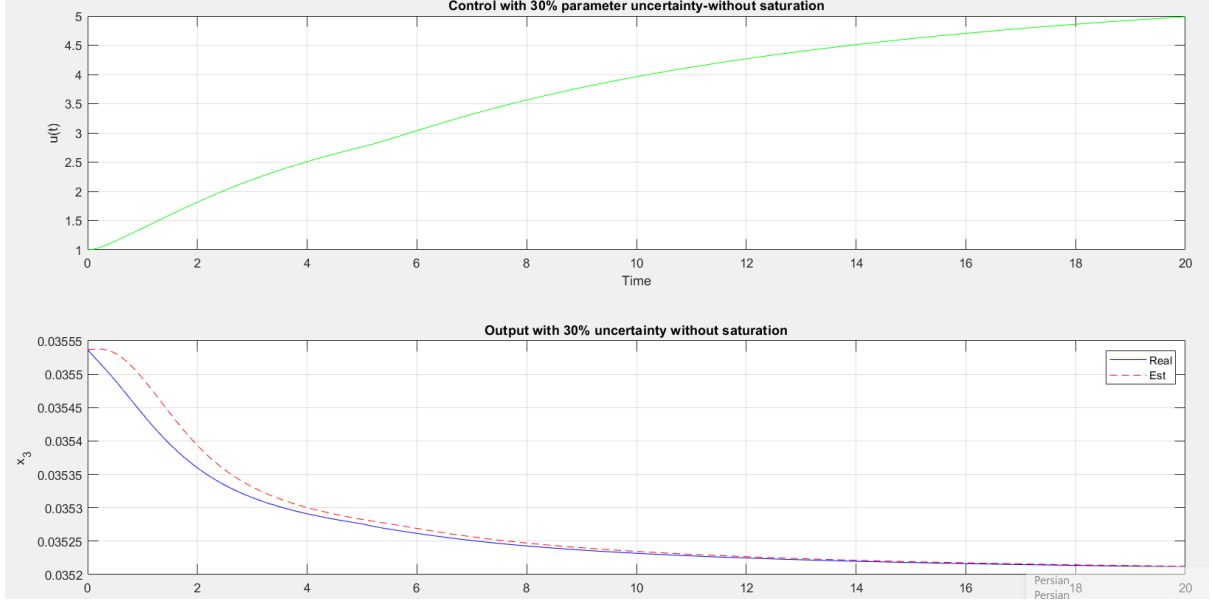


Figure 10: Robustness test with +30% in F_{os} (no saturation).

9 Conclusion

Phase 2 successfully completed the design and analysis of a complete control system for the CSTR reactor. The statefeedback controller exhibits a limited domain of attraction, which is mapped. The invariance of the transfer function across realizations is verified. Integral action eliminates steadystate error, but windup demands antiwindup measures. A PI observer correctly estimates the states even under disturbances, and its bandwidth can be tuned to balance speed and noise. The observerbased controller performs nearly as well as the ideal state feedback. Finally, the system tolerates a 30 % increase in the inlet flow rate, both with and without saturation, confirming its robustness.